

2N2222A

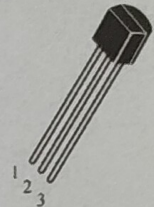
Package: TO-92

Maximum Ratings ($T_a = 25^\circ$)

Parameter	Symbol	Rating	Unit
Collector-base voltage	BV_{CBO}	75	V
Collector-emitter voltage	BV_{CEO}	40	V
Emitter-base voltage	BV_{EBO}	5	V
Collector current	I_{CM}	600	mA
Collector Power Dissipation	P_C	625	mW
Junction Temperature	T_j	150	$^\circ\text{C}$
Storage Temperature	T_{stg}	$-55 \sim +150$	$^\circ\text{C}$

TO-92

1. EMITTER
2. BASE
3. COLLECTOR



Electrical Characteristics ($T_a=25^\circ\text{C}$)

Parameter	Symbol	Test Condition	Min	Typ	Max	Unit
Collector-Base Breakdown Voltage	BV_{CBO}	$I_C=10\mu\text{A}$, $I_E=0$	75			V
Collector-Emitter Breakdown Voltage	BV_{CEO}	$I_C=10\text{mA}$, $I_B=0$	40			V
Emitter-Base Breakdown Voltage	BV_{EBO}	$I_E=10\mu\text{A}$, $I_C=0$	5			V
collector-base cut-off current	I_{CBO}	$V_{CB}=60\text{V}$, $I_E=0$			10	nA
collector-emitter cut-off current	I_{CEX}	$V_{CE}=40\text{V}$, $V_{EB}=3\text{V}$			10	nA
emitter-base cut-off current	I_{EBO}	$V_{EB}=3\text{V}$, $I_C=0$			10	nA
DC current gain	H_{FE}	$V_{CE}=10\text{V}$, $I_B=0.1\text{mA}$	35			
		$V_{CE}=10\text{V}$, $I_B=1\text{mA}$	50			
		$V_{CE}=10\text{V}$, $I_B=10\text{mA}$	75			
		$V_{CE}=10\text{V}$, $I_B=10\text{mA}$	35			
		$V_{CE}=10\text{V}$, $I_B=150\text{mA}$	100		300	
		$V_{CE}=1\text{V}$, $I_B=150\text{mA}$	50			
		$V_{CE}=10\text{V}$, $I_B=500\text{mA}$	40			
collector-emitter saturation voltage	V_{CESAT}	$I_C=150\text{mA}$, $I_B=15\text{mA}$			0.3	V
		$I_C=500\text{mA}$, $I_B=50\text{mA}$			1	
Base-Emitter Saturation Voltage	V_{BESAT}	$I_C=150\text{mA}$, $I_B=15\text{mA}$			1.2	V
		$I_C=500\text{mA}$, $I_B=50\text{mA}$			2	
Transition frequency	f_T	$V_{CE}=20\text{V}$, $I_B=20\text{mA}$, $f=100\text{MHz}$	300			MHz

Classification of H_{FE}

Rank	2N2222A-A	2N2222A-B
Range	100-200	200-300